



PC26 Objective 1.1C: Formulate and implement a stimulation roadmap to unlock productivity in Shah Simsima R1

Kick-Off Meeting

Andrey Semenov

Agenda



- I. PC26 Objective 1.1C description.
- II. Shah: Simsima R1 introduction.
- III. Shah: previous stimulation experience.
- IV. Framing session outcome.
- V. Shah 2026 planned treatments.
- VI. Candidate selection.
- VII. Objective 1.1C: SoW, deliverables, planning, governance.

Objective 1.1C: Formulate and implement a stimulation roadmap to unlock productivity in Shah Simsima R1



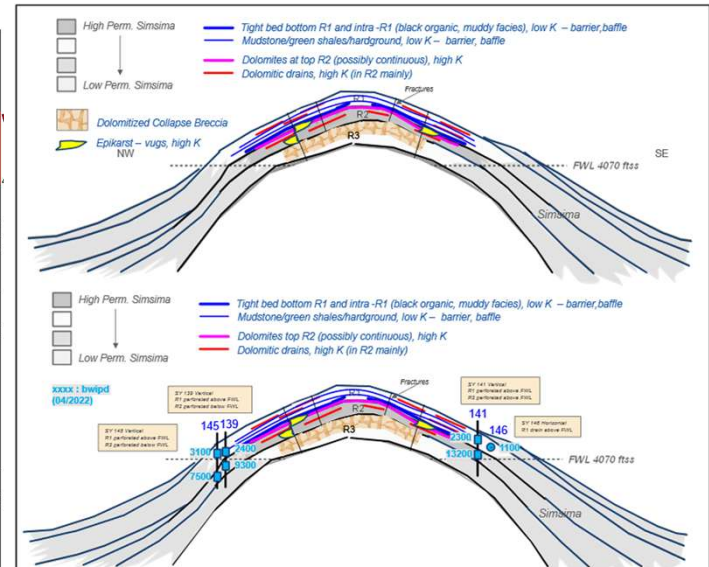
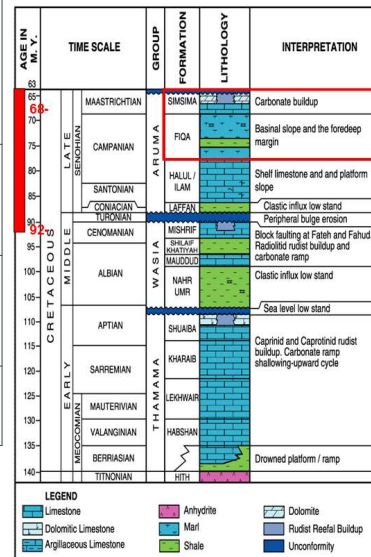
- Benchmark against global best practices and case studies from similar reservoirs to identify proven techniques.
- Design advanced stimulation methods for 5 wells by Q3,2026.

Shah Field Overview

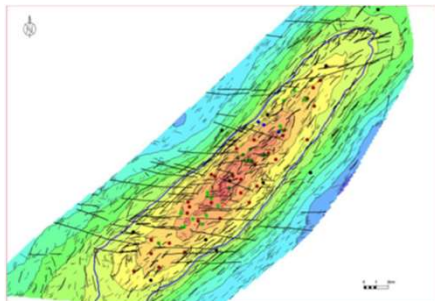
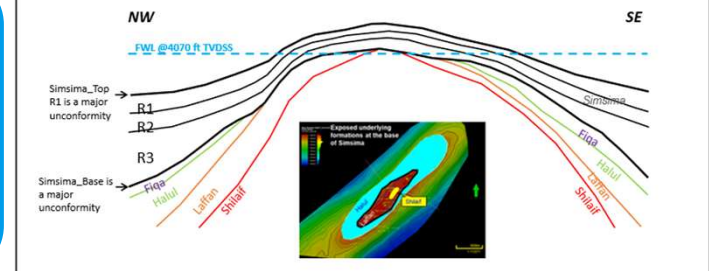


Discovery / FO	March 1966 (Shah1) / 1983
Structure	NE-SW faulted anticline, (flanks dip 4° - 7°) 6km x 26km / 160 km ² Complex carbonate fractured formation. Lateral and vertical lithological changes. Active aquifer support.
Development	1983 : Natural depletion - 17 producers - 23 Kbopd 1994: ESP implementation - increase to 50 kbopd 2014: Increase to sustainable rate 70 kbopd
Fluid	Undersaturated oil; Pb=392 psi, solution GOR 100 scf/stb, 30° API, oil density of 0.81 gm/cc and oil viscosity 2.3 cp at bubble point pressure
Major Projects	SHAH FDP Phase 1, FID 2008 & end execution Q3 2013 WI Pilot on-going, FFWI 2033

STOOIP = 3,439 MMSTB
NP 12/2022=590 MMBBL, RF=17.2%



Shah Simsima and underlying formations



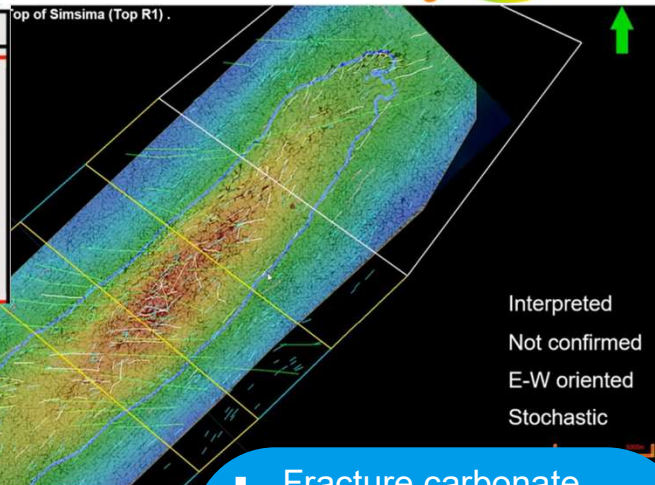
The only dual-media (fractured carbonate) development in ADNOC Onshore

Development: natural depletion, WI pilot, pressure support from edge regional aquifer appeared to be insufficient

Shah Simsima R1 reservoir



Facies/RQ Comments	Type Log (SY-06)	Reservoir Architecture	Mechanical Stratigraphy	Concepts
Peritidal cycles with org-rich tops, flat-shelled bivalves and some large rudist shells. Poor reservoir quality because organic material plugging pores Peritidal cycles with hardgrounds, borings and rootlets at the cycle top. Cycle bases are well-dolomitized with some dissolution of skeletal material creating the best reservoir quality. Peritidal cycles with org-rich tops. Reservoir quality poor because org. material plugging pores, impeding dolomitization. Prominent organic matter-rich layer at the base of R1		Thin-bedded. Base of peritidal cycles are the reservoir (dolomitized), tops are cemented or organic-rich. Best reservoir quality in the middle of R1. Top and Base are the poorest.	More prone to fracturing: Tight intervals, thin-bedded.	GR + K/Th and Dol. fraction logs to identify areas of less organic material and sweet spots of R1 Understand the type, number, and thickness of cycles across the reservoir Higher Dol frac to the South → better RQ Increase of GR to the North → poorer RQ



SUMMARY OF PRODUCTION PERFORMANCE YEARLY AVERAGE

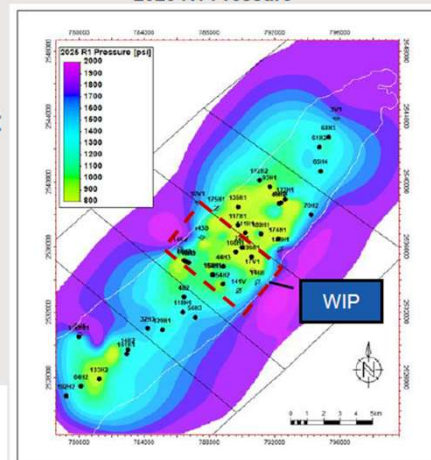
RTR (YA), MBD	Actual OP (YE), MBD	2025 FSOPR, MBD	Cum. Oil (YE), MMSTB	GOR (YE), SCF/STB	WC (YE), %	Total Producers (YE)	Active Producers (YE)
33	30.0	31.5	262.1	75	46.3	69*	63

*: 3 new OP drilled and waiting for Tie-in (SY-118, SY-192, SY-196)

KEY INSIGHTS AND HIGHLIGHTS:

- Actual oil production is slightly below plan, and the available TR capacity was insufficient to offset planned and unplanned losses.
- 2025 WA (5% of wells were off stream) improved compared to 2024
- Water cut has increased by 4% since 2024, reservoir pressure reduced wells' potential (low ESP intake and BHFP), and three new OP showed poor performance (-1 Mbd).
- By December 2025, 3 existing OP wells inactive (-1.5 Mbd)

2025 R1 Pressure



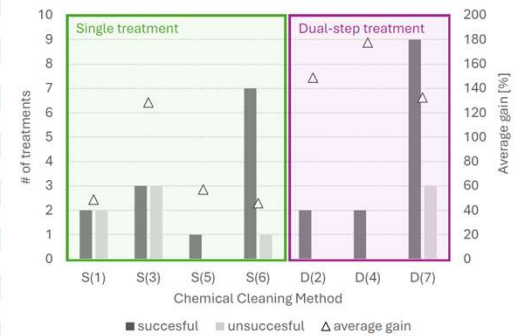
- Fracture carbonate reservoir, delivered 30 MBD in 2025 average production (46% WCT)
- Sector review: high WCT in the N/S and low reservoir pressure C/CN
- Structure: interpreted lineaments (curvature attribute) and stochastic fractures implemented

Shah: acid stimulation experience (SPE ADIPEC25)



Well	Date-i	WC2	P gain %	Q gain %	Y-tool	Method	Placement
17	4-Aug-24	25.0	180%	1150%	N	3	BH
1	1-Nov-24	24.4	32%	705%	N	3	BH
17	18-Jan-25	31.6	-59%	679%	N	7	BH
22	10-May-25	16.7	42%	335%	N	7	BH
18	3-Nov-24	23.0	97%	306%	Y	7	BH
24	22-Mar-24	3.5	0%	264%	Y	4	CT
10	26-May-24	41.8		235%	N	1	BH
3	5-Apr-24	61.5		196%	Y	2	CT
4	26-Mar-24	29.8		180%	Y	6	CT
5	21-May-25	67.9	125%	152%	N	7	BH
27	12-Jun-24	21.4	9%	102%	Y	2	CT
20	5-Jul-24	1.5	-18%	92%	Y	4	CT
23	7-Sep-24	1.1	6%	82%	Y	6	CT
17	1-Nov-23	26.7	84%	60%	N	1	BH
21	6-Jan-24	3.4	31%	57%	Y	5	CT
12	10-Jun-25	34.7	6%	51%	N	7	BH
13	4-Apr-25	46.1	53%	48%	Y	7	CT
11	7-May-24	22.4	-13%	43%	Y	7	BH
25	12-Dec-23	37.5	104%	43%	Y	6	CT
14	28-Jun-24	20.8		31%	Y	6	CT
16	2-Apr-25	22.7	29%	25%	Y	7	CT
19	7-Jun-24	12.5	0%	19%	Y	6	CT
9	3-Nov-24	34.5	0%	19%	N	3	BH
28	13-Mar-25	22.7	0%	18%	Y	7	CT
24	01-Sep-24	2.6	4%	9%	Y	6	CT
6	6-Jan-25	56.5		9%	Y	6	BH
26	2-May-25	40.4	6%	-4%	Y	7	CT
8	7-Sep-24	23.8	5%	-6%	Y	6	BH
2	8-Jun-25	70.7	31%	-6%	N	7	BH
7	2-Mar-24	33.1	-13%	-23%	N	3	BH
15	20-Mar-22	27.7	-4%	-25%	N	1	BH
12	2-Jul-23	35.9	-2%	-29%	N	3	BH
13	21-Dec-23	23.1	17%	-30%	Y	3	CT
14	19-Apr-25	40.9	86%	-56%	Y	7	CT
18	25-Mar-24	18.4	-35%	-75%	N	1	BH

Method	Company	Execution	Product
1	1	Single	Dispersant
2	1	Dual	(1) Dispersant, (2) acid
3	2	Single	Organic solvent
4	2	Dual	(1) Organic solvent, (2) acid
5	3	Single	Mutual solvent
6	4	Single	Naphtha
7	4	Dual	(1) Naphtha, (2) Acid

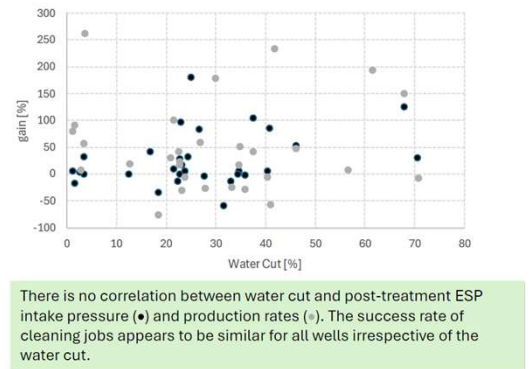


On average, the cleaning jobs executed in 2-steps were more successful than the single-step jobs

- single-step jobs: 1 in 3 jobs not successful
- dual-step jobs: 1 in 5 jobs not successful

No big difference between chemical placement method:

- Placement by CT: success date 76%, avg gain 125%
- Placement by BH: success date 72%, avg gain 140%



There is no correlation between water cut and post-treatment ESP intake pressure (●) and production rates (○). The success rate of cleaning jobs appears to be similar for all wells irrespective of the water cut.



Framing session



Following aspects of the treatment and stimulation process could be considered for further improvement:

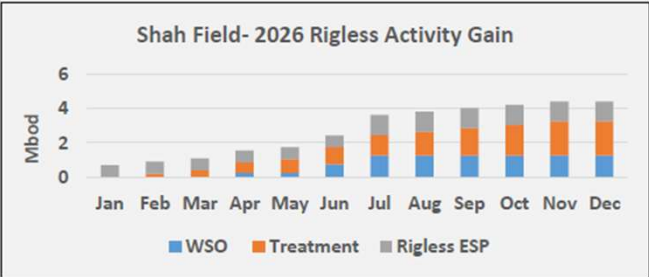
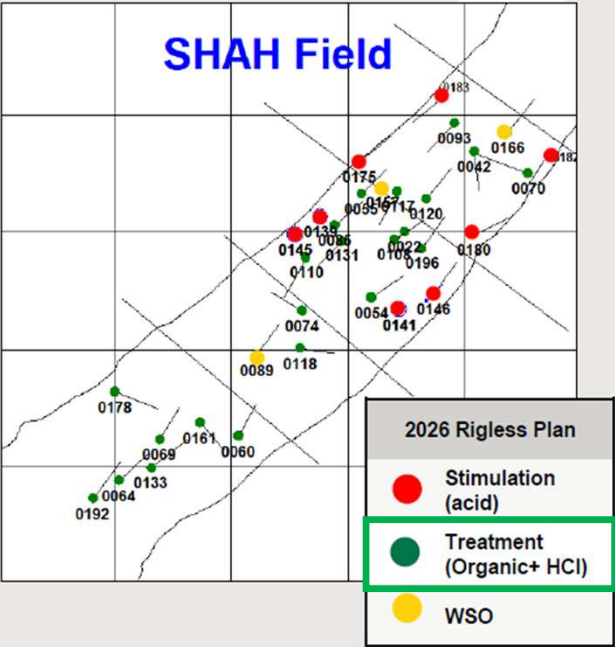
- Nafta and acid (15% HCl) 2 stage treatment have been considered the optimal based on experience. The possibility of single stage treatment (including both components) and improvement of acid composition (higher HCl concentration, different acids/recipe) to be considered.
- The acid to efficiently stimulate dolomite rock in Simsima R1.
- The stimulation treatment with removed upper completion (ESP, Y-tool, tubing) to be considered. In addition to nitrogen injection, available advanced activation methods could be utilized.
- Duration of the treatment effect and cost of the treatment to be taking into account.

Shah: ARPR25

2026 FIELD DEVELOPMENT ACTIVITIES - RIGLESS



Parameter	Planned for 2026	Comments
Total rigless activities	37	
WSO	3	1 Chemical WSO + 2 Mechanical WSO (SSD shifting)
Stimulation (acid)	12	Acid stimulation for injection strings (as needed) including 4 new WI (2 SAWI + 2 PWRI) part of commissioning SY-191 WDW
Treatment (dual step organic/inorganic)	20	Continue best practice of R1 wells treatment to improve productivity and ESP-Intake pressure (dual step treatment organic solvent/Naphtha and acid)
Rigless ESP- Cable deployment	2	Material available to recover temporary new ESP failures (SY-98 is in the plan)
Production/Injection gain from rigless activities, MBOPD/MBWPD	4	



Three WSO in R2 OP wells and Treatment plan for Sim R1 OP wells

Subject of the study

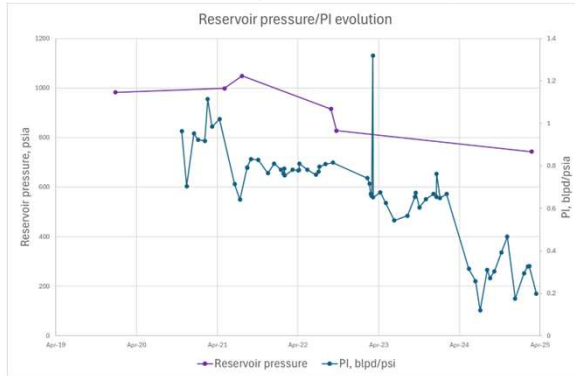


Shah Simsima R1: 5 candidates selection

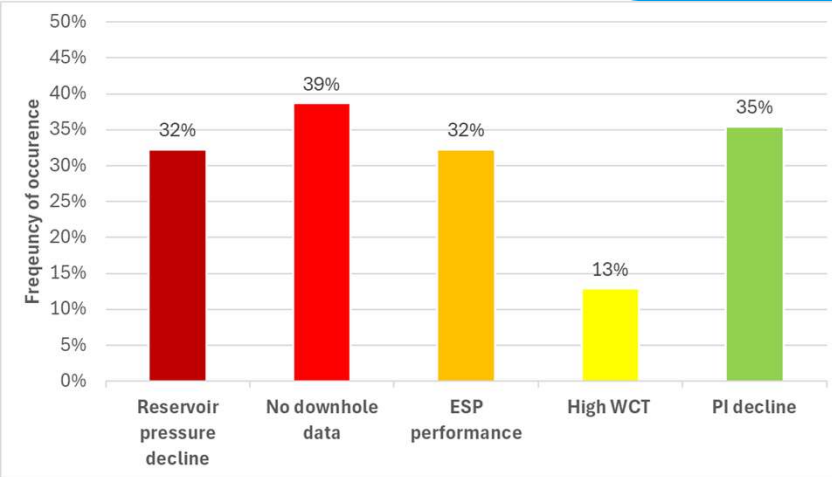


PI degradation (PC25 WP study): 2 wells

High dolomitization:
2 wells Simsima R1

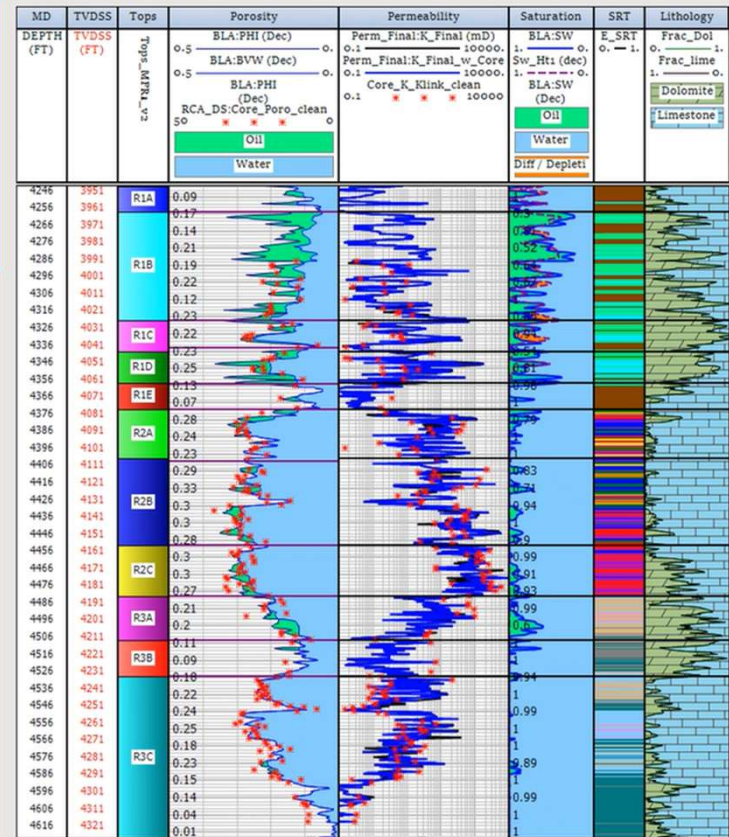


PI declined after the well start-up causing the moving of ESP to the left zone.
Examples: SY-0152, SY-0153, SY-0154, SY-0135, SY-0131



Water injector(poor injectivity): 1 well

SY-08



South-Est Obj. 1.1C: Formulate and implement a stimulation roadmap to unlock productivity in Shah Simsima R1

Obj.	Description		Q1	Q2	Q3	Q4
1.1C	- Benchmark against global best practices and case studies from similar reservoirs to identify proven techniques. - Design advanced stimulation methods for 5 wells by Q3,2026	SoW	- Scope of work and framing sessions - Data transfer to AL - Initial data QC	- Analysis of acid stimulation results (delivery and recipe) - Benchmarking vs analogs reservoir - Consider the possibility of treatment with retrieved upper completion	Selection of the optimal recipe and stimulation design for 5 wells	- Preparation of the draft of report - Integration of the comments from FP - Finalization of the report
		DELIVERABLES	- Initial data analysis - KOM - Selection of 5 wells candidates for the stimulation	Evaluation of different recipe and stimulation design	- Stimulation design and recommendations for 5 wells - WS: Shah Simsima R1 stimulation roadmap	- Close out session - Final report

Asset Lead FP	Asset FP
Andrey SEMENOV	Zahra Faisal Al Hashmi (ADNOC Onshore)
François WACHE	Mubarak Hulaiman
	Mubarak Alhameli (ADNOC Onshore)
	Hector Aguilar (ADNOC Onshore - SE-DEV)

Added value	Production, Growth
-------------	--------------------

The objective could be extended to 2027 (subject to PC27 signature) following the possibility to cover the following:

- Results of additional experiments on core plugs could add value, depends on the stimulation recommendations.
- Analysis of the treatment results recommended after execution by AON on candidate wells selected and additional monitoring actions performed.

Required data



- Stimulation and treatment reports on Shah Simsima (including chemical composition, design, results, etc..). Stimulation fluid preparation reports.
- Pre and post-stimulation performance results: well performance, PI evolution, PTA interpretation
- 5 wells candidates for the stimulation design:
 - End of well reports (including all information acquired during drilling and completion of the well)
 - Event log of all actions performed on the wells during the well life
 - Historical production, well test summary (MS Access DB Shah Simsima)
 - Transient well-test interpretation results (PI, etc...)
 - Shareholder guidelines



Governance

- Regular progress meetings.
 - Bi-weekly
- Workshops:
 - Q3: Shah Simsima R1 stimulation roadmap

